TinyBolus Formulary

Peer-review document

Version 0.0.75 · Updated 2026-11-04

Published by Aether Somnus

Developed by Sérgio Pinto, M.D.

Contact: tinybolusfeedback@gmail.com

Source: github.com/SGP-Prime/tinybolusformulas

Privacy policy: sgp-prime.github.io/tinybolus-site/privacy.html

---

This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.

About this document

This is the human-readable export of the TinyBolus paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

Age brackets

Bracket bounds are printed exactly as the application resolves them. A bound in *months* is inclusive in completed months ("≤12 mo" includes a child of exactly 1 y 0 m). A bound in *years* covers every child of that age in completed years ("1–6 y" runs through 6 y 11 m). Where a months bracket ends on a birthday, the next bracket starts one month later and is marked ">": ">1–6 y" begins at 1 y 1 m. "≥6 mo" starts at exactly 6 months. Weight bands are inclusive on both bounds.

Visibility

This document prints every row in the formulary. The app shows a curated default set and keeps the remaining rows behind a switch in Settings; those rows are tagged *hidden by default* beside their name below. A hidden row is still maintained, sourced and reviewed — it is simply off until a user enables it.

Hard floors

Some items carry an *age* or *weight* floor below which the formulary declines to suggest a dose. In the app this renders as ∅ (empty-set glyph) with a "not indicated" explanation; the same wording is used here. A floor records where the cited source stops (no data, not recommended, outside the licensed indication) — it is not a claim of formal contraindication. Items with such floors are flagged in red below each row.

References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

Source

The formulary itself lives at github.com/SGP-Prime/tinybolusformulas as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

Contents

1. EMERGENCY — PALS
2. Vital Signs
3. Induction
4. PONV
5. Painkillers
6. Emergency Drugs
7. Fluids and Blood
8. Equipment
9. Respiratory
10. Reversal Agents
11. Regional Anesthesia
12. Weight Estimation
13. Bibliography

EMERGENCY — PALS

Compression rate

VALUE `100–120/min`

References: [1]

Compression depth

RULE `4 cm <1 y · 5 cm ≥1 y`

not exceeding 6 cm at any age

References: [1]

C:V ratio

VALUE `15:2`

15:2 if paediatric-BLS trained · 30:2 otherwise

References: [1]

Defibrillation

DOSE `4 J/kg`

refractory VF/pVT (6th shock on): stepwise to 8 J/kg, max 360 J · adolescent 120–200 J

References: [1], [2]

Cardioversion (synchronised)

DOSE `1 J/kg`

double per attempt to max 4 J/kg

References: [1]

Adrenaline (cardiac arrest)

IV/IO

DOSE `0.01 mg/kg (max 1 mg)`

repeat every 3–5 min · max 1 mg/dose

References: [1], [3]

Amiodarone

IV/IO

DOSE `5 mg/kg (max 300 mg)`

after 3rd shock · repeat after 5th · max 150 mg subsequent

References: [1]

Atropine

IV

DOSE `0.02 mg/kg (max 0.5 mg)`

References: [1], [4], [5]

Adrenaline (bradycardia)

IV/IO

DOSE `1–2 µg/kg`

unresponsive to oxygenation/ventilation · atropine if vagal

References: [1], [4]

Adenosine

IV

DOSE `1st 0.1 mg/kg (max 6 mg) · 2nd 0.3 mg/kg (max 18 mg)`

2nd dose ≥1 min later · then q1–2 min, +0.05–0.1 mg/kg a dose, max 0.5 mg/kg · cardiology

References: [1]

Magnesium sulphate

HIDDEN BY DEFAULT

IV/IO

DOSE `50 mg/kg (max 2 g)`

References: [1]

Vital Signs

Heart Rate

≤1 mo	100–205 bpm
2–12 mo	100–180 bpm
>1–2 y	98–140 bpm
3–5 y	80–120 bpm
6–11 y	75–118 bpm
≥12 y	60–100 bpm

References: [6]

Systolic BP

≤1 mo	67–84 mmHg
2–12 mo	72–104 mmHg
>1–2 y	86–106 mmHg
3–5 y	89–112 mmHg
6–9 y	97–115 mmHg
10–11 y	102–120 mmHg
12–15 y	110–131 mmHg
≥16 y	90–140 mmHg

References: [6]

Diastolic BP

≤1 mo	35–53 mmHg
2–12 mo	37–56 mmHg
>1–2 y	42–63 mmHg
3–5 y	46–72 mmHg
6–9 y	57–76 mmHg
10–11 y	61–80 mmHg
12–15 y	64–83 mmHg
≥16 y	60–90 mmHg

References: [6]

Respiratory Rate

≤12 mo	30–53 /min
>1–3 y	22–37 /min
4–5 y	20–28 /min
6–12 y	18–25 /min
≥13 y	12–20 /min

References: [6]

Tidal Volume

DOSE `6–8 mL/kg`

per IDEAL body weight — if obese use estimated weight · lower end for neonates

References: [7], [8], [1]

Induction

Propofol

IV

≤1 mo	1–2.5 mg/kg
2–12 mo	3–4 mg/kg
>1–6 y	3–3.5 mg/kg
7–12 y	2.5–3 mg/kg
≥13 y	2–2.5 mg/kg

≈ halve after sevoflurane induction

References: [9], [10], [11], [12]

Thiopental

HIDDEN BY DEFAULT

IV

≤1 mo	3–4 mg/kg
2–12 mo	5–8 mg/kg
>1 y	5–6 mg/kg

then 1 mg/kg as needed · titrate to effect

References: [13]

Etomidate

HIDDEN BY DEFAULT

IV

DOSE `0.2–0.3 mg/kg`

for haemodynamic stability

References: [14], [15]

Ketamine

IV

DOSE `1–2 mg/kg`

References: [16], [14]

Ketamine (IM)

HIDDEN BY DEFAULT

IM

DOSE `4–5 mg/kg`

onset 3–5 min

References: [16], [14]

Fentanyl (induction)

IV

DOSE `1–3 µg/kg`

References: [17], [14], [18]

Midazolam

IV

≤6 mo	0.05–0.1 mg/kg — *<6 mo: prolonged effect · slow injection · hypotension with opioids*
7 mo – 5 y	0.05–0.1 mg/kg
≥6 y	0.025–0.05 mg/kg

CAP `max 5 mg`

References: [19], [14]

Midazolam (oral premed)

HIDDEN BY DEFAULT

PO

DOSE `0.5 mg/kg (max 20 mg)`

References: [14]

Midazolam (buccal premed)

HIDDEN BY DEFAULT

BUCCAL

DOSE `0.3 mg/kg (max 10 mg)`

if oral refused

References: [14]

Midazolam (IN premed)

HIDDEN BY DEFAULT

IN

DOSE `0.2–0.3 mg/kg (max 10 mg)`

5 mg/mL — divide between nostrils

References: [20], [14]

Dexmedetomidine (IN premed)

HIDDEN BY DEFAULT

IN

DOSE `1–3 µg/kg (max 200 µg)`

100 µg/mL solution · onset ~30 min

References: [21], [22]

Rocuronium

IV

DOSE `0.6–1.2 mg/kg`

upper range for RSI

References: [23], [24]

Succinylcholine

HIDDEN BY DEFAULT

IV

≤12 mo	2–3 mg/kg — *raised in infants — larger extracellular volume*
>1 y	1–2 mg/kg

intubating dose · avoid in malignant hyperthermia / hyperkalaemia risk

References: [24], [15], [25]

Atracurium

HIDDEN BY DEFAULT

IV

≤12 mo	0.3–0.4 mg/kg
>1 y	0.4–0.5 mg/kg

References: [14], [15]

PONV

Ondansetron

IV

DOSE `0.1 mg/kg (max 4 mg)`

Not indicated <1 months (age floor)

References: [26], [14], [27]

Dexamethasone (PONV)

IV

DOSE `0.15–0.25 mg/kg (max 4 mg)`

References: [26], [28], [14]

Droperidol (PONV)

HIDDEN BY DEFAULT

IV

DOSE `0.01–0.05 mg/kg (max 1.25 mg)`

Not indicated <24 months (age floor)

2nd line · avoid in long QT · give 30 min before end of surgery

References: [29], [30]

Painkillers

Paracetamol

IV

≤10 kg 7.5 mg/kg

>10 kg 15 mg/kg

CAP `max 1 g`

q6h (min 4 h) · max 4 doses/24 h

References: [31], [14], [32]

Ketorolac

IV

DOSE `0.5 mg/kg (max 30 mg)`

Not indicated <6 months (age floor)

References: [33], [14]

Morphine

IV

≤1 mo 0.025–0.05 mg/kg

≥2 mo 0.1 mg/kg

References: [34], [14]

Fentanyl (analgesia)

HIDDEN BY DEFAULT

IV

DOSE `0.5–1 µg/kg`

References: [14], [35], [18]

Metamizole

HIDDEN BY DEFAULT

IV

DOSE `8–16 mg/kg (max 1 g)`

Not indicated <3 months (age floor)

References: [36], [37]

Ibuprofen (oral)

HIDDEN BY DEFAULT

PO

DOSE `5–10 mg/kg (max 400 mg)`

Not indicated <3 months (age floor)

not <5 kg

References: [38], [14], [39]

Tramadol

HIDDEN BY DEFAULT

IV/PO

DOSE `1–2 mg/kg (max 100 mg)`

Not indicated <12 months (age floor)

q6h · respiratory depression caution

References: [40], [41]

Diclofenac (PO/PR)

HIDDEN BY DEFAULT

PO/PR

DOSE `0.3–1 mg/kg`

Not indicated <12 months (age floor)

q8–12h

References: [42], [41]

Emergency Drugs

Adrenaline (anaphylaxis IM)

IM

DOSE `0.01 mg/kg (max 0.5 mg)`

repeat q5 min as needed · auto-injectors: 0.15 mg 1–5 y · 0.3 mg 6–12 y · 0.5 mg >12 y

References: [1], [4]

Adrenaline (anaphylaxis IV)

IV

DOSE `1 µg/kg`

titrate to response · IM adrenaline is first-line

References: [1]

Adrenaline (anaphylaxis infusion)

HIDDEN BY DEFAULT IV

DOSE `0.5–1 mL/kg/h`

1 mg in 100 mL NaCl 0.9% = 10 µg/mL · dedicated pump line · after 2 failed IM doses

References: [43], [4]

Clemastine

IV

DOSE `0.025 mg/kg (max 2 mg)`

Not indicated <12 months (age floor)

second-line, after adrenaline

References: [44], [45], [46]

Atropine

IV

DOSE `0.02 mg/kg (max 0.5 mg)`

References: [1], [4], [5]

Ephedrine

IV

≤6 mo 0.6–1.2 mg/kg — *far higher than older-child dose – indirect agent, immature sympathetics*

≥7 mo 0.1–0.3 mg/kg

ampoule 30 mg/mL — dilute

References: [47], [14], [15]

Phenylephrine

IV

DOSE `1–20 µg/kg (max 500 µg)`

upper range for tet spells · hypotension: start low, titrate

References: [14], [15]

Lidocaine

IV

DOSE `1 mg/kg (max 100 mg)`

antiarrhythmic — refractory VF / pulseless VT

References: [1], [3]

Glucose 10%

IV/IO

DOSE `2 mL/kg`

hypoglycaemia · = 0.2 g/kg · recheck glucose after

References: [1], [4]

Glucagon

HIDDEN BY DEFAULT

IM/SC

≤25 kg 0.5 mg

>25 kg 1 mg

hypoglycaemia · no response after 10 min: repeat or IV glucose

References: [48]

Calcium gluconate 10%

IV/IO

DOSE `0.5 mL/kg (max 20 mL)`

hypocalcaemia · hyperkalaemia (ECG changes; not in arrest) · Ca-blocker overdose · slow IV

References: [1], [49]

Sodium bicarbonate 8.4%

IV/IO

DOSE `1 mmol/kg`

1 mL/kg of 8.4% · tricyclic toxicity / severe acidosis · not routine in arrest

References: [1], [49]

Dantrolene

IV

DOSE `2.5 mg/kg`

reconstitute NOW (volume per vial label) · repeat 1 mg/kg q5 min · may exceed 10 mg/kg

References: [50], [51]

Midazolam (buccal, status epilepticus)

HIDDEN BY DEFAULT

BUCCAL

≤11 mo	2.5 mg
1–4 y	5 mg
5–9 y	7.5 mg
≥10 y	10 mg

Not indicated <3 months (age floor)

seizure ≥5 min · 3–6 mo: hospital only · repeat once after 5–10 min

References: [52], [1]

Midazolam (IV, status epilepticus)

IV/IO

DOSE `0.15 mg/kg (max 10 mg)`

repeat once · off-label use

References: [53], [1]

Levetiracetam (status epilepticus)

HIDDEN BY DEFAULT

IV/IO

DOSE `40–60 mg/kg (max 4.5 g)`

Not indicated <1 months (age floor)

loading dose · 2nd line after 2 benzodiazepine doses · give over 5 min · off-label

References: [1], [53]

Insulin + glucose (hyperkalaemia)

IV

DOSE `0.1 units/kg (max 10 units)`

WITH glucose 10% 5 mL/kg (max 250 mL) — never insulin alone · check glucose & K+ q15min

References: [54], [1]

Salbutamol (nebulized, hyperkalaemia)

HIDDEN BY DEFAULT NEBULIZED

VALUE `2.5–5 mg`

2.5 mg ≤25 kg · 5 mg >25 kg · repeat ≤5× · with insulin–glucose

References: [54], [1]

Fluids and Blood

Fluid Bolus

IV

DOSE `10 mL/kg (max 500 mL)`

balanced crystalloid · cardiogenic shock: 5 mL/kg

References: [1], [4]

Maintenance

RULE `Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter`

4-2-1 (Holliday-Segar) · isotonic + glucose ± K+ · <1 mo: neonatal regimen

References: [55], [56]

Total Blood Volume

HIDDEN BY DEFAULT

≤1 mo	85 mL/kg
2 mo – 2 y	75 mL/kg
≥3 y	70 mL/kg

References: [15]

Tranexamic Acid

IV

DOSE `15 mg/kg (max 1 g)`

loading dose over 10 min · then 2 mg/kg/h for ≥8 h

References: [57], [58], [59], [1]

Max allowable blood loss

HIDDEN BY DEFAULT

VALUE `EBV × (Hct – min Hct) / Hct`

estimate, not a transfusion trigger · blood volume ~70–100 mL/kg · source uses mean Hct

References: [60], [61]

Equipment

ETT

RULE `Cuffed: 3.5 + age/4, rounded to ±0.5`

<1 MO `Neonatal weight table (≤4 kg, uncuffed; heavier neonates use the infant age table):`

≤1.5 kg · preterm ≤32 wk	size 2.5
≤3 kg · 33–39 wk	size 3.0
≤4 kg · term ≥39 wk	size 3.5

INFANT `age table (cuffed):`

≤6 mo	size 3.5
7–12 mo	size 4.0

cuffed sizes — uncuffed add 0.5 (≥1 y) · neonatal table: uncuffed sizes

References: [62], [63]

ETT insertion depth

RULE `6 + weight (<1 mo) · weight/2 + 8 (<1 y) · age/2 + 12 (≥1 y)` *(cm)*

nasal: +1 cm <1 y · +2–3 cm ≥1 y

References: [64], [65], [49]

---

iGel LMA

IGEL SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–12 kg	size 1.5
10–25 kg	size 2
25–35 kg	size 2.5
30–60 kg	size 3
50–90 kg	size 4
90–∞ kg	size 5

References: [66], [67]

AuraGain LMA

HIDDEN BY DEFAULT

AURAGAIN SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–10 kg	size 1.5
10–20 kg	size 2
20–30 kg	size 2.5
30–50 kg	size 3
50–70 kg	size 4
70–100 kg	size 5
100–∞ kg	size 6

References: [68]

Laryngoscope blade

≤1 y	Miller 0–1
2–7 y	Macintosh 2
≥8 y	Macintosh 3

Miller lifts the epiglottis · Macintosh 1 equally effective

References: [69], [15]

Face mask

≤1 mo	Size 0
2–12 mo	Size 1
>1–3 y	Size 2
4–8 y	Size 3
9–14 y	Size 4
≥15 y	Size 5

cover nose to chin cleft — anatomical fit over size

References: [15], [70]

Reservoir bag

≤6 mo	0.5 L
7 mo – 6 y	1 L
≥7 y	2 L

References: [71], [72]

Suction catheter

HIDDEN BY DEFAULT

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

References: [49], [15]

Nasogastric / orogastric tube

HIDDEN BY DEFAULT

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

larger sizes for adolescent decompression

References: [49], [15]

Oral / nasal airway sizing

HIDDEN BY DEFAULT

VALUE `by landmark`

Guedel: incisors → jaw angle · nasal: nostril → tragus −1 cm; not in basal-skull fracture

References: [73], [74], [75]

Respiratory

Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

100 µg/puff · repeat incrementally as needed

References: [76], [77]

Salbutamol (nebulized)

HIDDEN BY DEFAULT NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

References: [76], [77]

Ipratropium (nebulized)

HIDDEN BY DEFAULT NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [14]

Magnesium sulphate (asthma)

IV

DOSE `40 mg/kg (max 2 g)`

infuse over 20 min

References: [78], [14], [1]

Adrenaline (nebulized)

NEBULIZED

DOSE `0.5 mg/kg (max 5 mg)`

croup / stridor · 1:1000 (1 mg/mL)

References: [79], [14]

Dexamethasone (anti-inflammatory)

HIDDEN BY DEFAULT

IV

DOSE `0.5 mg/kg (max 8 mg)`

airway oedema / post-extubation stridor

References: [80], [81]

Propofol (laryngospasm)

IV

DOSE `0.5 mg/kg`

give in increments

References: [82]

Succinylcholine (laryngospasm)

IV

DOSE `0.1–0.5 mg/kg`

lower end may spare breathing · avoid in malignant hyperthermia / hyperkalaemia risk

References: [83], [82]

Succinylcholine (IM)

IM

DOSE `4 mg/kg (max 200 mg)`

laryngospasm or no IV access · avoid in malignant hyperthermia / hyperkalaemia risk

References: [83], [82]

Rocuronium (laryngospasm)

IV

DOSE `0.6–1.2 mg/kg`

when succinylcholine contraindicated (malignant hyperthermia) — then secure the airway

References: [24], [23]

Salbutamol (IV, severe asthma)

HIDDEN BY DEFAULT

IV

DOSE `0.015 mg/kg (max 0.25 mg)`

bolus over 10 min · infusion 1–2 µg/kg/min (>2: ICU) · monitor K+ · off-label <12 y

References: [76], [84]

Hydrocortisone (IV, asthma)

HIDDEN BY DEFAULT

IV

DOSE `4 mg/kg`

q4h — if oral not tolerated

References: [76]

Reversal Agents

Naloxone (titrated)

IV

DOSE `0.01 mg/kg`

titrate to respiration · repeat q2–3 min

References: [14], [1]

Naloxone (overdose)

IV/IO/IM

DOSE `0.1 mg/kg (max 2 mg)`

References: [1], [14]

Flumazenil

IV

DOSE `0.01 mg/kg (max 0.2 mg)`

repeat ≤4× · total max 0.05 mg/kg (up to 1 mg) · licensed ≥1 y

References: [14]

Sugammadex

IV

DOSE `2–4 mg/kg`

upper dose for deep block

References: [85], [86]

Sugammadex (profound block)

HIDDEN BY DEFAULT

IV

DOSE `16 mg/kg`

can't intubate / can't oxygenate: reverses full-dose rocuronium · off-label in children

References: [87], [88]

Neostigmine

HIDDEN BY DEFAULT

IV

DOSE `0.04–0.07 mg/kg (max 5 mg)`

with atropine · slow over 1–2 min, after first recovery signs

References: [14], [15]

Atropine (with neostigmine)

HIDDEN BY DEFAULT IV

DOSE `0.02 mg/kg (max 0.5 mg)`

with or just before neostigmine

References: [14], [15]

Neostigmine + glycopyrronium

HIDDEN BY DEFAULT IV

DOSE `0.02 mL/kg (max 2 mL)`

References: [89]

Regional Anesthesia

Lidocaine (perineural)

PERINEURAL

≤5 mo maximum 3.5 mg/kg

≥6 mo maximum 5 mg/kg

plain solution — no adrenaline

References: [15], [14]

Bupivacaine (perineural)

PERINEURAL

≤5 mo maximum 1.5–2 mg/kg

≥6 mo maximum 2–2.5 mg/kg

infusion: 0.1–0.3 mg/kg/h

References: [90], [91]

Ropivacaine (perineural)

PERINEURAL

≤5 mo maximum 2 mg/kg

≥6 mo maximum 3 mg/kg

infusion: 0.1–0.3 mg/kg/h

References: [92], [93], [91]

Intralipid 20% bolus

IV

DOSE `1.5 mL/kg`

LA toxicity rescue · repeat if unstable · max total 12 mL/kg

References: [90], [94]

Intralipid 20% infusion

IV

DOSE `0.25 mL/kg/min`

LA toxicity rescue – continue ≥15 min after stability

References: [90], [94]

Levobupivacaine (max)

PERINEURAL

DOSE `maximum 1.25 mg/kg (max 150 mg)`

Not indicated <6 months (age floor)

SmPC figure: ilioinguinal block, per side · no data <6 mo

References: [95]

Weight Estimation

Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

1–12 mo (infant)

FORMULA `weight (kg) = (0.5 × months) + 4`

References: [49]

1–5 y (young child)

FORMULA `weight (kg) = (2 × age) + 8`

References: [49]

6–12 y (school-age)

FORMULA `weight (kg) = (3 × age) + 7`

References: [96], [49]

Bibliography

Citations numbered in order of first encounter throughout this document.

1. Djakow J, Turner NM, Skellett S, Buysse CMP, Cardona F, de Lucas N, del Castillo J, Kiviranta P, Lauridsen KG, Markel F, Martinez-Mejias A, Roggen I, Biarent D. European Resuscitation Council Guidelines 2025: Paediatric Life Support. Resuscitation 2025;215(Suppl 1):110767.
2. Acworth J, Del Castillo J, Tiwari LK, et al. Energy doses for paediatric defibrillation in cardiac arrest: systematic review and meta-analysis. Resuscitation Plus 2025;24:100991.
3. Lasa JJ, Dhillon GS, Duff JP, et al. Part 8: Pediatric Advanced Life Support: 2025 American Heart Association and American Academy of Pediatrics Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation. 2025;152(16_suppl_2):S479–S537.
4. Resuscitation Council UK. Paediatric emergency algorithms & resources (version 6, February 2026; reflects ERC Guidelines 2025). resus.org.uk.
5. Barrington KJ. The myth of a minimum dose for atropine. Pediatrics 2011;127(4):783–784.
6. American Heart Association. Pediatric Advanced Life Support (PALS) Provider Manual: normal heart rate, respiratory rate and blood pressure by age (tables; neonatal BP row = 'neonate, 96 h'). Dallas, TX: AHA; 2016 (2015 Guidelines edition).
7. Spaeth J, Schumann S, Humphreys S. Understanding pediatric ventilation in the operative setting. Part II: Setting perioperative ventilation. Paediatr Anaesth 2022;32(2):247–254.
8. Kneyber MCJ, de Luca D, Calderini E, et al. Recommendations for mechanical ventilation of critically ill children from the Paediatric Mechanical Ventilation Consensus Conference (PEMVECC). Intensive Care Med 2017;43(12):1764–1780.
9. Aun CST, Short SM, Leung DH, Oh TE. Induction dose-response of propofol in unpremedicated children. Br J Anaesth 1992;68(1):64–67.
10. Westrin P. The induction dose of propofol in infants 1–6 months of age and in children 10–16 years of age. Anesthesiology 1991;74(3):455–458.
11. van Dijk H, Hendriks MP, van Eck-Smaling MM, van Wolfswinkel L, van Loon K. Age-stratified propofol dosage for pediatric procedural sedation and analgesia. Anesth Analg 2023;136(3):551–558.
12. Kim SH, Hong JY, Suk EH, Jeong SM, Park PH. Optimum bolus dose of propofol for tracheal intubation during sevoflurane induction without neuromuscular blockade in children. Anaesth Intensive Care 2011;39(5):899–903.
13. Mercury Pharmaceuticals (ADVANZ). Thiopental powder for solution for injection · Summary of Product Characteristics (paediatric induction: newborns 3–4, infants 5–8, children 5–6 mg/kg). electronic medicines compendium, medicines.org.uk/emc/product/665 (current edition).
14. Paediatric Formulary Committee. BNF for Children (online). London: BMJ Group, Pharmaceutical Press, and RCPCH Publications.
15. Cote CJ, Lerman J, Anderson BJ, eds. A Practice of Anesthesia for Infants and Children, 6th edition. Philadelphia: Elsevier; 2018.
16. Craven R. Ketamine. Anaesthesia 2007;62(Suppl 1):48–53.
17. Fentanyl 50 micrograms/ml Solution for Injection/Infusion - Summary of Product Characteristics (children 2–11 years: initial 1–3 micrograms/kg). electronic medicines compendium,

medicines.org.uk/emc/product/6617 (current edition).

18. Cravero JP, Agarwal R, Berde C, et al. The Society for Pediatric Anesthesia recommendations for the use of opioids in children during the perioperative period. Paediatr Anaesth 2019;29(6):547–571.

19. Midazolam 5 mg/ml solution for injection/infusion · Summary of Product Characteristics (sedation: 6 months–5 years initial 0.05–0.1 mg/kg; 6–12 years 0.025–0.05 mg/kg; <6 months: conscious sedation not recommended). electronic medicines compendium, medicines.org.uk/emc/product/15365 (current edition).

20. Baldwa NM, Padvi AV, Dave NM, Garasia MB. Atomised intranasal midazolam spray as premedication in pediatric patients: comparison between two doses of 0.2 and 0.3 mg/kg. J Anesth 2012;26(3):346–350.

21. Yuen VM, Hui TW, Irwin MG, Yao TJ, Chan L, Wong GL, Hasan MS, Shariffuddin II. A randomised comparison of two intranasal dexmedetomidine doses for premedication in children. Anaesthesia 2012;67(11):1210–1216.

22. Jun JH, Kim KN, Kim JY, Song SM. The effects of intranasal dexmedetomidine premedication in children: a systematic review and meta-analysis. Can J Anaesth 2017;64(9):947–961.

23. Taivainen T, Meretoja OA, Erkola O, Rautoma P, Juvakoski M. Rocuronium in infants, children and adults during balanced anaesthesia. Paediatr Anaesth 1996;6(4):271–275.

24. Mazurek AJ, Rae B, Hann S, et al. Rocuronium versus succinylcholine: are they equally effective during rapid-sequence induction of anesthesia? Anesth Analg 1998;87(6):1259–1262.

25. Meakin G, McKiernan EP, Morris P, Baker RD. Dose-response curves for suxamethonium in neonates, infants and children. Br J Anaesth 1989;62(6):655–658.

26. Gan TJ, Belani KG, Bergese S, et al. Fourth Consensus Guidelines for the Management of Postoperative Nausea and Vomiting. Anesth Analg 2020;131(2):411–448.

27. hameln pharma ltd. Ondansetron 2 mg/ml solution for injection - Summary of Product Characteristics (PONV: children ≥1 month, 0.1 mg/kg IV max 4 mg). electronic medicines compendium, medicines.org.uk/emc/product/6469 (last updated 2023; product since discontinued).

28. Madan R, Bhatia A, Chakithandy S, Subramaniam R, Rammohan G, Deshpande S, Singh M, Kaul HL. Prophylactic dexamethasone for postoperative nausea and vomiting in pediatric strabismus surgery: a dose ranging and safety evaluation study. Anesth Analg 2005;100(6):1622–1626.

29. Panpharma. Droperidol 2.5 mg/mL solution for injection (paediatric PONV, second line: 10–50 µg/kg, max 1.25 mg, not <2 y) - Summary of Product Characteristics. electronic medicines compendium (medicines.org.uk, current edition).

30. Nederlands Kenniscentrum Farmacotherapie bij Kinderen. Kinderformularium: droperidol (PONV, 2–18 y: 10–50 µg/kg/dose, max 1.25 mg/dose). kinderformularium.nl (current edition).

31. Anderson BJ. Paracetamol (acetaminophen): mechanisms of action. Paediatr Anaesth 2008;18(10):915–921.

32. Paracetamol 10 mg/mL solution for infusion — Summary of Product Characteristics. electronic Medicines Compendium (emc), current edition.

33. Gupta A, Daggett C, Ludwick J, Wells W, Lewis A. Ketorolac after congenital heart surgery: does it increase the risk of significant bleeding complications? Paediatr Anaesth 2005;15(2):139–142.

34. Bouwmeester NJ, Anderson BJ, Tibboel D, Holford NH. Developmental pharmacokinetics of morphine and its metabolites in neonates, infants and young children. Br J Anaesth 2004;92(2):208–217.

35. Association of Paediatric Anaesthetists of Great Britain and Ireland. Good Practice in Postoperative and Procedural Pain Management, 2nd edition. Paediatr Anaesth 2012;22(Suppl 1):1–79.

36. Nederlands Kenniscentrum Farmacotherapie bij Kinderen. Kinderformularium: metamizol (IV, 3 months–18 years: 8–16 mg/kg/dose, max 4 doses/day, max 1 g/dose). kinderformularium.nl/geneesmiddel/40460/metamizol (current edition).

37. Fieler M, Eich C, Becke K, et al. Metamizole for postoperative pain therapy in 1177 children: a prospective, multicentre, observational, postauthorisation safety study. Eur J Anaesthesiol 2015;32(12):839–843.

38. Southey ER, Soares-Weiser K, Kleijnen J. Systematic review and meta-analysis of the clinical safety and tolerability of ibuprofen compared with paracetamol in paediatric pain and fever. Curr Med Res Opin 2009;25(9):2207–2222.

39. Reckitt Benckiser Healthcare (UK) Ltd. Nurofen for Children Orange (3 months to 12 years) · Summary of Product Characteristics. electronic medicines compendium, medicines.org.uk/emc/product/396 (current edition).

40. AS Kalceks. Tramadol 50 mg/ml solution for injection/infusion · Summary of Product Characteristics (children ≥1 y: 1–2 mg/kg/dose; daily max 8 mg/kg or 400 mg). electronic medicines compendium, medicines.org.uk/emc/product/13177 (current edition).

41. Nederlands Kenniscentrum Farmacotherapie bij Kinderen. Kinderformularium (Dutch national paediatric formulary). kinderformularium.nl (current edition).

42. Novartis. Voltarol 12.5 mg Suppositories · Summary of Product Characteristics (juvenile idiopathic arthritis 1–12 y: 1–3 mg/kg/day divided). electronic medicines compendium, medicines.org.uk/emc/product/1044 (current edition).

43. Resuscitation Council UK. Emergency treatment of anaphylaxis: guidelines for healthcare providers (incl. the refractory anaphylaxis algorithm). May 2021. resus.org.uk.

44. Ring J, Beyer K, Biedermann T, et al. Guideline (S2k) on acute therapy and management of anaphylaxis: 2021 update. Allergo J Int 2021;30(1):1–25.

45. Tavegyl (clemastine fumarate) 1 mg/mL solution for injection — Summary of Product Characteristics, current edition.

46. Nederlands Kenniscentrum Farmacotherapie bij Kinderen. Kinderformularium: clemastine (25–50 µg/kg/dose IV/IM, max 2 mg/dose). kinderformularium.nl (current edition).

47. Szostek AS, Saunier C, Elsensohn MH, Boucher P, Merquiol F, Gerst A, et al. Effective dose of ephedrine for treatment of hypotension after induction of general anaesthesia in neonates and infants less than 6 months of age: a multicentre randomised, controlled, open label, dose escalation trial. Br J Anaesth 2023;130(5):603–610.

48. Nederlands Kenniscentrum Farmacotherapie bij Kinderen. Kinderformularium: glucagon (severe hypoglycaemia: <25 kg 0.5 mg, >25 kg 1 mg SC/IM; repeat or give IV glucose if no effect after 10 min). kinderformularium.nl/geneesmiddel/40104/glucagon (current edition).

49. Advanced Life Support Group. Advanced Paediatric Life Support: A Practical Approach to Emergencies, 7th edition. Chichester: Wiley-Blackwell; 2023.

50. Hopkins PM, Girard T, Dalay S, Jenkins B, Thacker A, Patteril M, McGrady E. Malignant hyperthermia 2020: Guideline from the Association of Anaesthetists. Anaesthesia 2021;76(5):655–664.

51. Glahn KPE, Ellis FR, Halsall PJ, Müller CR, Snoeck MMJ, Urwyler A, Wappler F. Recognizing and managing a malignant hyperthermia crisis: guidelines from the European Malignant Hyperthermia Group. Br J Anaesth 2010;105(4):417–420.

52. Takeda. Buccolam (midazolam) oromucosal solution · Summary of Product Characteristics (prolonged acute convulsive seizures, 3 months to <18 years). electronic medicines compendium, medicines.org.uk/emc/product/2768 (current edition).

53. Resuscitation Council UK. Treating convulsive status epilepticus in children (paediatric emergency algorithms, version 6, February 2026; reflects ERC Guidelines 2025). resus.org.uk (current edition).

54. Royal Children's Hospital Melbourne. Clinical Practice Guideline: Hyperkalaemia. rch.org.au/clinicalguide (current edition).

55. Holliday MA, Segar WE. The maintenance need for water in parenteral fluid therapy. Pediatrics 1957;19(5):823–832.

56. Feld LG, Neuspiel DR, Foster BA, et al. Clinical Practice Guideline: Maintenance Intravenous Fluids in Children. Pediatrics 2018;142(6):e20183083.

57. Royal College of Paediatrics and Child Health. Evidence statement: major trauma and the use of tranexamic acid in children. RCPCH; November 2012.

58. Faraoni D, Goobie SM. The efficacy of antifibrinolytic drugs in children undergoing noncardiac surgery: a systematic review of the literature. Anesth Analg 2014;118(3):628–636.

59. Goobie SM, Meier PM, Pereira LM, et al. Efficacy of tranexamic acid in pediatric craniosynostosis surgery: a double-blind, placebo-controlled trial. Anesthesiology 2011;114(4):862–871.

60. Gross JB. Estimating allowable blood loss: corrected for dilution. Anesthesiology. 1983;58(3):277–280.

61. Breglio AM, Eck JB. Allowable Blood Loss (Question of the Week #176). Society for Pediatric Anesthesia, 2023. pedsanesthesia.org.

62. Duracher C, Schmautz E, Martinon C, et al. Evaluation of cuffed tracheal tube size predicted using the Khine formula in children. Paediatr Anaesth 2008;18(2):113–118.

63. European Resuscitation Council. ERC Guidelines 2025: Newborn resuscitation and support of transition of infants at birth. Resuscitation. 2025;215(Suppl 1):110766.

64. Phipps LM, Thomas NJ, Gilmore RK, et al. Prospective assessment of guidelines for determining appropriate depth of endotracheal tube placement in children. Pediatr Crit Care Med 2005;6(5):519–522.

65. Lau N, Playfor SD, Rashid A, Dhanarass M. New formulae for predicting tracheal tube length. Paediatr Anaesth 2006;16(12):1238–1243.

66. Intersurgical Ltd. i-gel User Guide: Adult and Paediatric Sizes. Intersurgical, Wokingham, UK (current edition).

67. Beringer RM, Kelly F, Cook TM, et al. A cohort evaluation of the paediatric i-gel airway during anaesthesia in 120 children. Anaesthesia 2011;66(12):1121–1126.

68. Ambu A/S. Ambu AuraGain Single Use Laryngeal Mask — Instructions for Use. V7, August 2023.

69. Passi Y, Sathyamoorthy M, Lerman J, Heard C, Marino M. Comparison of the laryngoscopy views with the size 1 Miller and Macintosh laryngoscope blades lifting the epiglottis or the base of the tongue in infants and children <2 yr of age. Br J Anaesth 2014;113(5):869–874.

70. Rendell-Baker L, Soucek DH. New pediatric face masks and anaesthetic equipment. Br Med J 1962;1(5293):1690.

71. Kalra A. Pediatric Anesthesia Digital Handbook. Department of Anesthesiology, Tufts Medical Center, Boston, MA (maskinduction.com), current edition.

72. Intersurgical Ltd. Reservoir bags (paediatric 0.5 L and 1 L; adult 2 L; 3 L also available). Intersurgical, Wokingham, UK (current edition).

73. Van de Voorde P, Turner NM, Djakow J, et al. European Resuscitation Council Guidelines 2021: Paediatric Life Support. Resuscitation. 2021;161:327–387.

74. Nemeth M, et al. Resuscitation. 2023;184:109702 (GUEDEL-I: oropharyngeal airway sizing by facial landmarks; PMID 36702339).

75. Nemeth M, et al. Resuscitation. 2021 (WEND:LI: nasopharyngeal airway sizing landmarks; PMID 34600970).

76. British Thoracic Society / Scottish Intercollegiate Guidelines Network. British guideline on the management of asthma (SIGN 158). Edinburgh: BTS/SIGN; 2019, updated 2024.

77. Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention, 2024 update. https://ginasthma.org/

78. Griffiths B, Kew KM. Intravenous magnesium sulfate for treating children with acute asthma in the emergency department. Cochrane Database Syst Rev 2016;4:CD011050.

79. Bjornson C, Russell K, Vandermeer B, Klassen TP, Johnson DW. Nebulized epinephrine for croup in children. Cochrane Database Syst Rev 2013;(10):CD006619.

80. Anene O, Meert KL, Uy H, Simpson P, Sarnaik AP. Dexamethasone for the prevention of postextubation airway obstruction: a prospective, randomized, double-blind, placebo-controlled trial. Crit Care Med 1996;24(10):1666–1669.

81. Malhotra D, Gurcoo S, Qazi S, Gupta S. Randomized comparative efficacy of dexamethasone to prevent postextubation upper airway complications in children and adults in ICU. Indian J Anaesth 2009;53(4):442–449.

82. Gavel G, Walker RWM. Laryngospasm in anaesthesia. Contin Educ Anaesth Crit Care Pain 2014;14(2):47–51.

83. Larson CP Jr. Laryngospasm — the best treatment. Anesthesiology 1998;89(5):1293–1294.

84. Resuscitation Council UK. Paediatric Life Support (basic and advanced) Guidelines 2025 (published 27 October 2025). resus.org.uk.

85. Plaud B, Meretoja O, Hofmockel R, et al. Reversal of rocuronium-induced neuromuscular blockade with sugammadex in pediatric and adult surgical patients. Anesthesiology 2009;110(2):284–294.

86. Lee S, Chung W. Sugammadex for our little ones: a brief narrative review. Anesth Pain Med (Seoul) 2024;19(4):269–279.

87. Lee C, Jahr JS, Candiotti KA, Warriner B, Zornow MH, Naguib M. Reversal of profound neuromuscular block by sugammadex administered three minutes after rocuronium: a comparison with spontaneous recovery from succinylcholine. Anesthesiology 2009;110(5):1020–1025.

88. Bridion (sugammadex) prescribing information. Merck Sharp & Dohme LLC (current revision).

89. Martindale Pharma. Glycopyrronium Bromide 0.5 mg / Neostigmine Metilsulfate 2.5 mg per ml solution for injection · Summary of Product Characteristics (paediatric 0.02 mL/kg IV; total dose not to exceed 2 mL). electronic medicines compendium, medicines.org.uk/emc/product/3390 (current edition).

90. Neal JM, Barrington MJ, Fettiplace MR, et al. The third American Society of Regional Anesthesia and Pain Medicine practice advisory on local anesthetic systemic toxicity: executive summary 2017. Reg Anesth Pain Med 2018;43(2):113–123.

91. Suresh S, Ecoffey C, Bosenberg A, et al. The European Society of Regional Anaesthesia and Pain Therapy/American Society of Regional Anesthesia and Pain Medicine recommendations on local anesthetics and adjuvants dosage in pediatric regional anesthesia. Reg Anesth Pain Med 2018;43(2):211–216.

92. Lönnqvist PA, Westrin P, Larsson BA, et al. Ropivacaine pharmacokinetics after caudal block in 1–8-year-old children. Br J Anaesth 2000;85(4):506–511.

93. Bosenberg AT, Thomas J, Lopez T, et al. Plasma concentrations of ropivacaine following a single-shot caudal block of 1, 2 or 3 mg/kg in children. Acta Anaesthesiol Scand 2001;45(10):1276–1280.

94. American Society of Regional Anesthesia and Pain Medicine. Checklist for treatment of local anesthetic systemic toxicity. ASRA Pain Medicine; 2020 revision.

95. Altan. Levobupivacaine 2.5 mg/ml · Summary of Product Characteristics (children: max 1.25 mg/kg per side, ilioinguinal/iliohypogastric; no data <6 months). electronic medicines compendium, medicines.org.uk/emc/product/13642 (current edition).

96. Luscombe MD, Owens BD, Burke D. Weight estimation in paediatrics: a comparison of the APLS formula and the formula 'Weight = 3(age)+7'. Emerg Med J 2011;28(7):590–593.